

Multiple Choice (10 marks)

1. Let $I = (S,V)$ be a MAC. Suppose $S(k,m)$ is always 5 bits long. Can this MAC be secure?
 - (a) No, an attacker can simply guess the tag for messages
 - (b) It depends on the details of the MAC
 - (c) Yes, the attacker cannot generate a valid tag for any message
 - (d) Yes, the PRG is pseudorandom
2. A sender must not be able to deny sending a message that was sent, is known as
 - (a) Message Nonrepudiation
 - (b) Message Integrity
 - (c) Message Confidentiality
 - (d) Message Sending
3. Which Nmap scan does not completely open a TCP connection?
 - (a) SYN stealth scan
 - (b) TCP connect
 - (c) XMAS tree scan
 - (d) ACK scan
4. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused
5. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet

True / False (10 marks)

Answer True or False

6. True or False: WPA 2 is more secure than WPA 3 due to its longer encryption keys and advanced features.
7. True or False: Wireless traffic sniffing is a process used for analyzing wireless traffic in forensic investigations and troubleshooting.
8. True or False: Authorization aims to restrict what operations and data a user can access within a system.
9. True or False: The four primary security principles related to messages include Confidentiality, Access Control, Integrity, and Non-repudiation.
10. True or False: Message authentication primarily focuses on protecting the confidentiality of the message content during transmission.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the perimeter of a pentagon.
12. Write a function to find all pairs in an integer array whose sum is equal to a given number.

End of Paper